Living Lab Workshop 2026

HPC and LLMs

Apurv D. Kulkarni

Contents

- HPC overview
- LLM Access
  - TUD:AI LLM Services
  - Self-deployment using HPC job
- Using Self-deployed LLM Service with HPC

HPC Overview

- Compute Nodes
  - Barnard
  - Romeo
  - Julia
  - Alpha
  - Capella

GPU Nodes

Where AI models can run faster

- CLI and Acess via Jupyter Notebook

	Alpha	Capella
Hardware	A100	H100
Nodes	35*	142*
GPU/Node	8	4
Mem/GPU	40 GB	94 GB

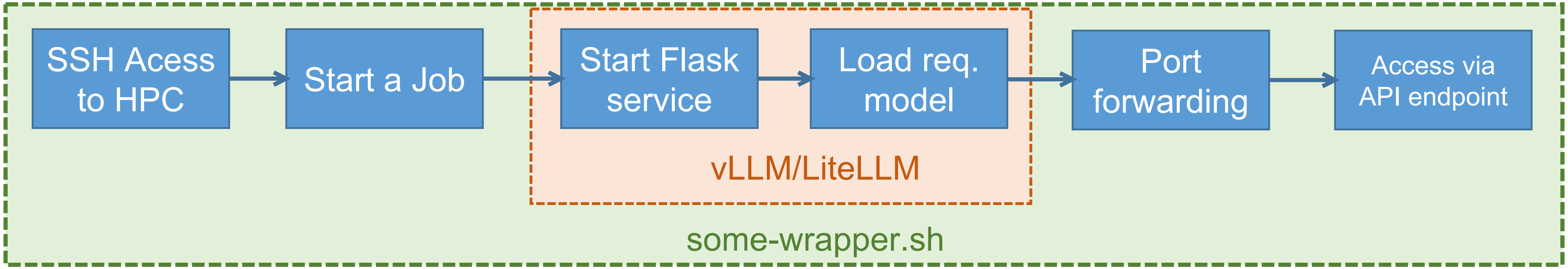
- Filesystem: Specialized storage where designed for optimized IO
  - `cat`
  - `horse`
  - `quokka`

LLM Hosting

- TUD:AI / Academic cloud
  - Easy access using API key
  - Optimized for faster inference
  - Not all models are available
  - Documentation: https://llm.scads.ai/docs/
- Self-Hosting
  - Any model can be hosted
  - Stable avalability
    - Long term
    - Downtime independent
  - Can be a little tricky
  - Requires knowledge of HPC hardware and software infra.

LLM Hosting: Self-Hosting

- Idea
  - We can‘t just run a separate job for every inference
  - Need to host the model as a service via Flask/FastAPI

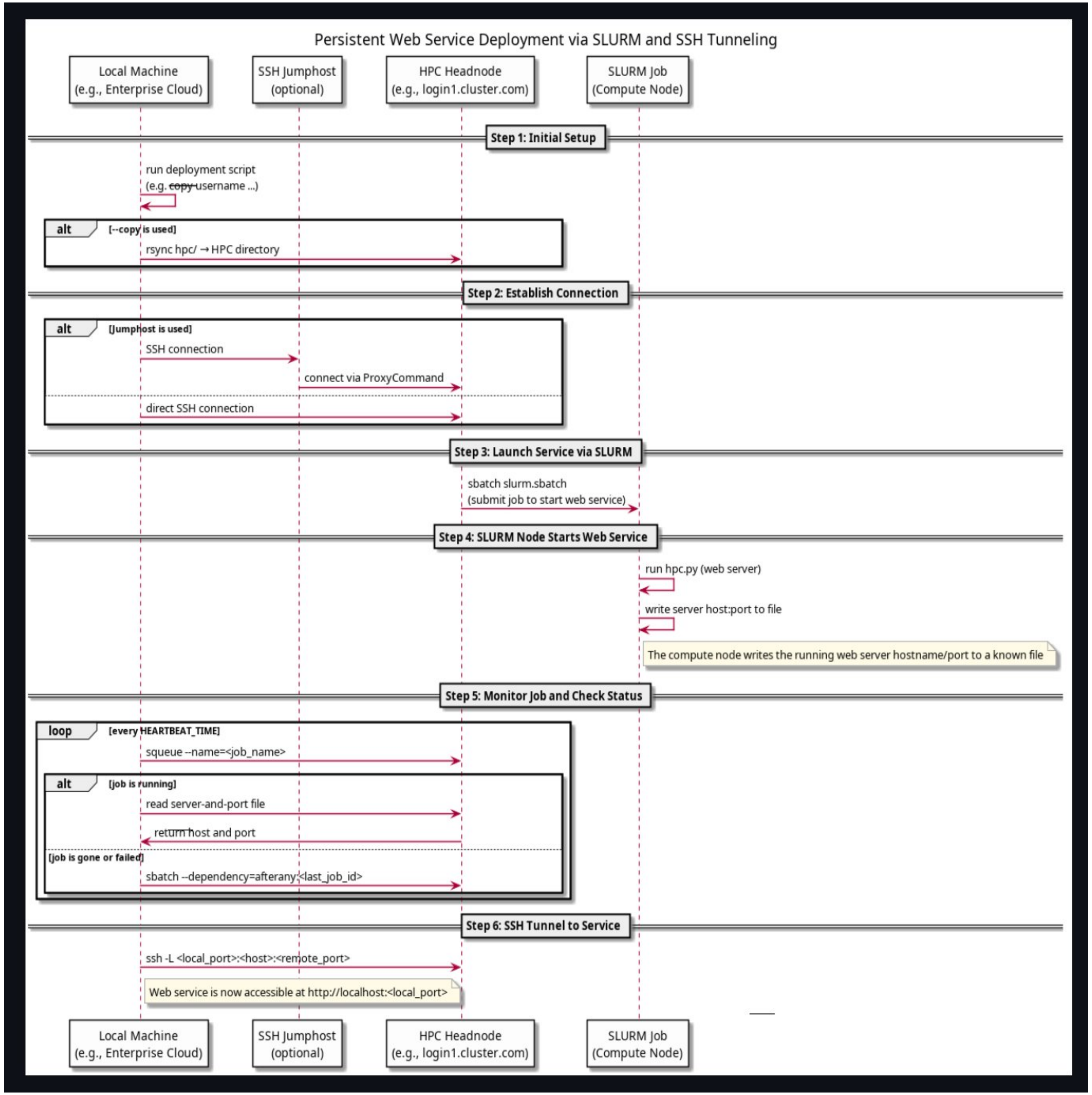


- Requirements
  - Which hardware ? And how many resources ?
  - Which Modules (software) ?
  - A bash script - automating all the steps
  - An sbatch template – running required steps to load/serve the model

Persistent Web Services on SLURM

- Enables you to deploy and persist a web-based service (e.g., an API to a resource-intensive Job)
- Automatically monitors the service job, restarts it if necessary
- Forwards the connection securely to your local machine (even across jumphosts)

https://github.com/NormanTUD/hpcservice



For Our Workshop

- Model - OmniVoice
  - Not available on TUD:AI
- Setting up and starting a job using script
  - Clone: https://github.com/apurvkulkarni7/LivingLab-Team-Retreat-2026.git
  - Go to: `LivingLab-Team-Retreat-2026/retreat/10-deployment/hpc-llm/omnivoice-tts`
  - Setup configuration (using provided example): `omnivoice.conf`
  - Run launch script: `./launch_omnivoice.sh start`

```
 ./launch_omnivoice.sh start

==> Connecting to HPC
[ OK ]  SSH connection to login2.alpha.hpc.tu-dresden.de

==> Syncing server code
[INFO]  Syncing server code to /data/horse/ws/lilahpc-test

==> Syncing model
[INFO]  model 'omnivoice' is a Hugging Face model (downloaded on the cluster); skipping rsync

==> Installing deps
[INFO]  Installing vLLM-Omni on the cluster (first run may take a while)...
```

Already Setup

For Our Workshop

- Open a terminal and simply type

```
ssh -L 8000:<NodeID>:8000 lilahpc@login1.alpha.hpc.tu-dresden.de
```

- What it does ?
  - Connects your laptop/machine to already running instance of model service
- Once conncected, run:

```
python3 client/speech_client_stdlib.py --text "Hello, how are you?"
```

or

```
curl -X POST http://localhost:8000/v1/audio/speech   \
         -H 'Content-Type: application/json'   \
         -d '{"input": "Hello, how are you?","language":"English","response_format":"wav"}' \
         -o out.wav
```